

6.2 Linear combinations of random variables

Probability and Statistics 2 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Linearity gives $E(aX+bY)=aE(X)+bE(Y)$. For independent X,Y , $\text{Var}(aX+bY)=a^2\text{Var}(X)+b^2\text{Var}(Y)$; covariance changes this if dependence exists. Variances are not subtracted when finding difference of independent variables.

Worked example: follow the reasoning

If independent X,Y each have variance 4, $\text{Var}(X-Y)=4+4=8$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Independent X,Y have $E(X)=3$, $E(Y)=5$, $\text{Var}(X)=2$, $\text{Var}(Y)=4$. Find $E(2X-Y)$ and $\text{Var}(2X-Y)$. [3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

$E=2(3)-5=1$ [1]; $\text{Var}=4(2)+4=12$ [2].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.